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Economic effects of environmental quality change on recreational hunting in northwestern Saskatchewan: a contingent behaviour analysis

K.M. Morton, W.L. Adamowicz, and P.C. Boxall

Abstract: This study estimates changes in the value of a recreational hunting experience as road access, game populations, congestion, and travel distance change in the forest environment because of harvesting operations. Values were estimated using a model that considers changes in one or more environmental quality factors simultaneously. The model is a variant of the contingent valuation method that evaluates multiple environmental quality changes. The model is applied to big-game hunters in a particular forest management licence area in northern Saskatchewan. The results suggest that the welfare of these hunters increases with reductions in congestion and improvements in game populations and access levels. The annual value of those improvements that generate the highest welfare for moose and deer hunters ranges from about \$12 000 to \$19 000, depending on the welfare measure used. The highest values of welfare losses ranged from about -\$2750 to -\$19 500 depending on the welfare measure, and were associated with increases in congestion and decreases in access and game population levels. The methods and results presented in this paper suggest an approach for the values of nontimber products, such as recreational hunting, to be incorporated more explicitly in integrated forest management plans.

Résumé : Cette étude évalue les changements dans la valeur des expériences de chasse récréative en forêt, en réponse aux changements dans l'accès routier, les populations de gibiers, la congestion et les distances à parcourir associés aux opérations forestières. Les valeurs ont été calculées à l'aide d'un modèle qui tient compte simultanément de la variation d'un ou de plusieurs facteurs environnementaux. Ce modèle est une variante de la méthode d'évaluation contingente (« contingent valuation »), qui évalue des changements multiples de qualité environnementale. Ce modèle a été appliqué à des chasseurs de grande faune dans une zone de licence d'aménagement forestier du nord de la Saskatchewan. Les résultats suggèrent que le bien-être de ces chasseurs s'améliore lorsque la congestion baisse et la taille et l'accès des populations de gibiers augmentent. La valeur annuelle de ces augmentations qui génèrent le bien-être le plus élevé des chasseurs d'orignal et de cerf varie d'environ 12 000 \$ à 19 000 \$, selon la mesure de bien-être utilisée. Les plus grandes pertes de bien-être allaient d'environ -2 750 \$ à -19 500 \$, selon la mesure de bien-être, et étaient associées à l'augmentation de la congestion et à la baisse de la taille et de l'accès des populations de gibiers. Les méthodes et résultats présentés dans cet article suggèrent une approche d'incorporation plus explicite de l'évaluation des produits autres que la matière ligneuse, tels que la chasse récréative, dans les plans d'aménagement forestier intégré.

[Traduit par la Rédaction]

Introduction

Integrated resource management (IRM) espouses a philosophy that considers managing forests for more than fibre supplies for lumber or pulp production and suggests that management regimes be developed that maximize

societal benefits (Loomis 1993). For an IRM plan to be successfully designed, social values for nontimber components of forests must be recognized and monetary values be placed on nontimber goods and services to facilitate their comparison with timber values in planning and management scenarios. There are, however, no readily available monetary valuations for a majority of nontimber resources because they are not bought and sold in markets.

There are techniques developed for estimating market values for nonmarket goods and for assessing changes in their values due to environmental improvements or degradation. These techniques use one of two classes of methodologies. The first uses an indirect approach that involves observing an individual's behaviour and then modelling

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the behaviour in an economic framework. The most common indirect method is the travel cost model. The second utilizes a direct approach that usually involves surveys, administered in person or by mail, to determine how economic decisions are made, and which uses this knowledge to value a particular good or service. The most common direct method of valuing nonmarket goods is called contingent valuation (CV). A typical CV study involves the use of surveys or interviews to create a hypothetical market in which an individual's value is estimated by asking for their willingness to pay (WTP) for the good or their willingness to accept compensation for loss of the good (Mitchell and Carson 1989).

The purpose of this study is to assess the impacts of changes in a set of recreation-quality factors that are in turn influenced by timber harvesting operations. A model is developed specifically for big-game hunting and is used to value potential changes in game populations, road access, hunter congestion, and travel costs that result from forest industry activities. Knowledge of these impacts allows the examination of trade-offs between various forest management actions, and thus contributes to the choice of an IRM plan that includes an assessment of societal benefits.

A variation of the CV method is used to value the non-market good – recreational big-game hunting² in this study. This approach is called contingent behaviour (CB), and it differs from the traditional CV method in that survey respondents are asked whether they would be willing to change their behaviour in response to price or quality changes. The respondents are presented with a choice between hunting in a base area (with attributes very much like present hunting areas in northeastern Saskatchewan) or hunting in an area with slightly different attributes (with at least one attribute improved in this case). The latter area, however, will cost the hunters more in terms of additional travel costs. Note that this is a “per-trip” choice occasion and the behaviour being modelled is site choice, not the number of trips. This type of question exactly parallels a CV question of the following form: “Would you be willing to pay \$X.XX in additional costs for an improvement in the following hunting quality attributes (game populations, access improvements, etc)?” This type of model has been used in the CV literature (e.g., Loomis et al. 1989) and has recently been employed as a parallel to discrete choice travel cost methods (Adamowicz et al. 1994). The differences between this approach and the one most CV studies follow are (1) quality attributes are varied over hunters and (2) the choice is presented as a choice between two areas rather than a willingness to pay for an improvement. Except for these differences in presentation structure, this choice process follows exactly the discrete choice CV model. The key is that this CB model is valuing the improvements to a base case in terms of willingness to pay additional travel costs. The costs figures are equivalent to bid amounts that would appear in CV studies. In other words,

individuals are not asked, “Would you be willing to pay \$X to hunt in a new zone?” Rather, they are asked whether they would be willing to visit an improved hunting site that has an additional cost per trip.

Respondents are asked to choose one of two forest hunting scenarios in which the first is a “base case” representing current forest conditions with associated hunting quality variables and the second represents altered levels of various qualities in the forest due to timber harvesting. The second site, however, also requires the hunter to pay additional travel costs to reach it. The choice of this “improved” site reflects a WTP at least \$X for the quality improvements, where X is the amount of additional cost the hunter is asked to incur. Using techniques common in the CV literature, this choice information can be used to develop a measure of expected WTP for the quality improvements. The impetus for developing this CB method was to examine how individuals trade off differing levels of multiple environmental qualities. This approach may outperform the traditional CV approach (Mitchell and Carson 1989) because the choice between sites may be less hypothetical than asking for a WTP for changes in environmental quality.

Theoretical background

The basis of the CB model is random utility theory. This involves modelling choices based on the notion that a hunter chooses the option that maximizes his or her utility, but that random elements influence this choice. An individual's utility can be represented as the sum of a systematic and a random component:

$$[1] \quad U_{in} = V(z_{in}, S_n) + \varepsilon(z_{in}, S_n) = V_{in} + \varepsilon_{in}$$

where U represents overall utility, V is the systematic portion of utility, z_{in} is a vector of site attributes, S_n is a vector of attributes associated with the individual (e.g., income), and ε represents the random portion. The subscripts i and n index choices (hunting sites) and individuals, respectively.

The probability that individual n will choose alternative i is equal to the probability that the utility received from i is greater than or equal to the utility received from any other alternative, represented by j , in the possible set of choices, C , faced by n . This is portrayed by

$$[2] \quad \Pr(i|C_n) = \Pr[U_{in} \geq U_{jn}, \text{ all } j \in C_n] \\ = \Pr[V_{in} + \varepsilon_{in} \geq V_{jn} + \varepsilon_{jn}, \text{ all } j \in C_n]$$

In the CB analysis reported here, an individual is faced with exactly two alternatives to choose from (i = zone A or zone B). The random utility model can then be developed into a binary choice model where the individual must choose between two hunting zones: zone A and zone B. The dependent variable, y , takes on the following values, where zone B represents an altered environment with additional travel costs and zone A represents current conditions or the base state:

$$[3] \quad y_{in} = \begin{cases} 0 & \text{if person } n \text{ chose zone A} \\ 1 & \text{if person } n \text{ chose zone B} \end{cases}$$

The probabilities of an individual choosing zone A or B can be written as

² Hunting is a nonmarket good for Canadian residents because in the vast majority of cases, hunters do not purchase hunting trips in a true market. Hunting licence fees are paid, but the fee is set in an arbitrary manner by the government. Licences are not sold in a market.

Fig. 1. An example of the contingent behaviour question used to examine the economic significance of changes in access, numbers of big game, and congestion to hunters in Saskatchewan. Note that the \$ X in the last line of the description of zone B was a cost figure sampled from a uniform distribution of costs bounded by \$1.00 and \$50.00. A similar question was used for moose hunters.

Consider the choice of hunting in sites A or B:

A. The following description is a representation of a typical forest stand in the mixed forests of northwestern Saskatchewan.

- The forest trails in this area are easily passable in dry weather in a 2-wheel drive vehicle, but in foul or wet weather, access is difficult even with a 4-wheel drive vehicle.
- On a typical day you will see, or find evidence (tracks, scrapes, rubs or droppings) of, **6 to 10 whitetail deer**.
- Your hunting party **will not encounter** another hunting party on your trip.

B. The following description represents a similar forest stand in the mixed forest of northwestern Saskatchewan.

- The roads into this forest are maintained by the forest products company during harvesting and are a mixture of sand and clay (no gravel) that are easily passable in dry weather in a 2-wheel drive vehicle. In foul or wet weather, access is difficult, even with a 4-wheel drive vehicle.
- Because of harvesting and replanting, there is increased vegetation for species like whitetail deer to eat.
- On a typical day you may see, or find evidence (tracks, scrapes, rubs or droppings) of, **8 to 12 whitetail deer**.
- Your hunting party **will encounter** another hunting party unfamiliar to you on your trip.
- In order to hunt in this area, you will have to travel farther and it will cost YOU an extra \$ X to get there.

In which zone would you prefer to hunt?

☐ **A**

☐ **B**

$$[4] \quad \Pr(\text{zone A}) = \Pr(y = 0) = \Pr(\varepsilon_{1n} - \varepsilon_{0n} \leq V_{0n} - V_{1n})$$

$$\Pr(\text{zone B}) = \Pr(y = 1) = \Pr(\varepsilon_{0n} - \varepsilon_{1n} \leq V_{1n} - V_{0n})$$

Linear utility functions of the following form are usually specified for the systematic component:

$$[5] \quad V_{in} = \beta' z_{in} + \gamma_i' S_n$$

where β and γ are vectors of unknown parameters. Assuming that each random term is type I extreme value distributed (and that the difference between random terms is logistically distributed (Maddala 1983)) the probability that individual n chooses alternative i is given by

$$[6] \quad \Pr(y_{in} = i) = \frac{\exp(V_{in} - V_{jn})}{1 + \exp(V_{in} - V_{jn})}$$

Incorporating the deterministic component of the V_{in} 's from [5], [6] becomes

$$[7] \quad \Pr(y_{in} = i) = \frac{\exp[\beta'(z_{in} - z_{jn}) + (\gamma_i - \gamma_j)' S_n]}{1 + \exp[\beta'(z_{in} - z_{jn}) + (\gamma_i - \gamma_j)' S_n]}$$

This formulation corresponds to a binary logit model estimated on the difference between attributes of the alternatives and the levels of individual characteristics. The binary logit model can be estimated using maximum likelihood techniques (Ben-Akiva and Lerman 1985).

Welfare theory

It is necessary to determine whether hunters will be better or worse off, in terms of welfare, in the post-logging scenario. The binary choice model presented above can be used to estimate a value of the change in economic welfare associated with the value of recreational hunting in zone B. One approach to this, suggested by Hanemann (1984) and Cooper and Loomis (1992), is to estimate the mean or

expected WTP. In the case where negative WTP amounts are ruled out a priori, this involves calculating

$$[8] \quad E(WTP) = \int_0^{\infty} [1 - \Pr(WTP \leq P)] dP$$

If the indirect utility function, V_{in} , takes on the following linear form (where P is the price or travel cost):

$$[9] \quad V_i = \beta_0 + \beta_1(P) + \beta_2(z_2) + \dots + \beta_K(z_K) + \gamma_1(S_1) + \dots + \gamma_M(S_M)$$

and

$$[10] \quad \alpha = \beta_0 + \sum_{k=2}^K \beta_k \bar{z}_k + \sum_{m=1}^M \gamma_m \bar{S}_m$$

then the mean WTP is calculated as

$$[11] \quad P^+ = \frac{1}{-\beta_1} \ln(1 + e^{\alpha})$$

A second approach used by Hanemann (1984) and Cooper and Loomis (1992) involves calculating the median of the distribution of WTP, or equivalently, the mean WTP given that WTP can take on negative values and given a linear model. Since these measures are equivalent for the linear model, we refer to both as the median WTP. The median WTP can also be thought of as the cost of going to zone B "when the individual is just at the point of indifference" between going to zone A and zone B. In other words, there is a 50:50 chance that the individual would be willing to incur the extra cost to visit zone B. The median WTP (or $E(WTP)$ given the possibility of negative WTP values) can be calculated as

$$[12] \quad P^- = \frac{\alpha}{-\beta_1}$$

Table 1. A description of hunting site attributes, their levels, and presentation in contingent behaviour questions in surveys sent to big-game hunters in Saskatchewan.

Attribute	Attribute levels						
	All survey versions, Zone A	Survey version 1		Survey version 2		Survey version 3	
		Zone B Q.1 ^a	Zone B Q.2	Zone B Q.1	Zone B Q.2	Zone B Q.1	Zone B Q.2
Presence of forestry operations ^b	No	Yes	Yes	Yes	Yes	Yes	Yes
Access ^c	Limited	Limited	Improved	Improved	Improved	Limited	Improved
Size of game populations ^d	Low	Increased	Low	Low	Increased	Increased	Increased
Congestion ^e	Low	Increased	Increased	Low	Increased	Low	Low
Cost ^f	No	Yes	Yes	Yes	Yes	Yes	Yes

^aEach survey contained 2 contingent behaviour questions: question 1 (Q.1) and question 2 (Q.2).

^bThe description for zone A does not indicate to the respondent that any forestry operations take place. The description for zone B stated that harvesting and replanting occurs in the zone and that a forestry company maintains the roads during harvesting.

^cLimited access is described as being easily passable in dry weather in a two-wheel drive vehicle, but in foul or wet weather, access is difficult even with a four-wheel drive vehicle. Improved access is the same as the limited access with the addition of old forest trails that are also passable with a two-wheel drive vehicle.

^dLow whitetail deer populations were described as a hunter seeing or finding evidence (tracks, scrapes, rubs, or droppings) of 6 to 10 deer on a typical day. Increased deer populations were described as seeing or finding evidence of 8 to 12 deer during a typical day. Moose populations were presented in the same way, with low described as seeing or finding evidence (tracks, droppings, rubs, or wallows) of 2 moose on a typical day, and increased being evidence of 3 or 4 moose.

^eLow congestion is defined as not encountering another hunting party during a trip. Increased congestion was defined as encountering another hunting party unfamiliar to the respondent during his or her trip.

^fZone A did not have a cost factor associated with hunting. Zone B, however, had a randomly generated cost included as an attribute. This attribute was presented to the respondent such that in order to hunt in zone B, you will have to travel farther and it will cost him or her an extra amount to get there.

where estimates for α and β are derived from maximum likelihood estimation of [7].

A third approach to welfare measurement in this type of model is the calculation of the marginal value of each attribute. The marginal value of an attribute (z_k) is the WTP for a unit change in an attribute, with everything else held constant. For the model presented above, the marginal values are β_k/β_1 , for $k = 2 \dots K$.

Methods

Data collection

Data used to develop and test the model came from mail surveys of whitetail deer (*Odocoileus virginianus*) and moose (*Alces alces*) hunters in Saskatchewan. Samples of hunters were obtained from hunting licences sold to residents hunting in the 1991 hunting season. A mail-back questionnaire was developed and sent to 1602 deer hunters and 1546 moose hunters. To help maximize the response rates for the surveys, the total design method developed by Dillman (1978) was utilized. Response rates were 51.6% for the moose hunters and 59.3% for the deer hunters. Morton et al. (1993) describe the development of the questionnaire and summarize some general descriptive results.

Three parts of the survey are relevant to the objectives of this study. The first involved respondents rating on a 5-point Likert scale the importance of factors they consider

in selecting hunting areas. These involved factors such as familiarity, access, distance, proximity to friends and relatives, and encounters with other hunters. The second involved a question where respondents were asked to rate 11 environmental and social factors thought to influence enjoyment of a hunting trip. These factors included various types of road closures, increased access, observing types of wildlife, encountering direct evidence of timber harvesting such as slash or replanted forest areas, and hunter congestion. Ratings of these factors involved a 5-point Likert scale, where 1 was "lessens enjoyment" and 5 was the highest level of "increases enjoyment." The third part of the survey involved the CB questions. Each survey contained two CB questions where three site characteristics relevant to hunting were varied. The characteristics were road quality or access, size of game populations, and hunter congestion. These characteristics were determined during discussions with big game hunters in a focus group held before the study. An example of this question is shown in Fig. 1.

Levels of the characteristics (Table 1) were varied such that six different site-choice questions were derived. Since hunters were presented with two questions in each questionnaire, three versions of both the whitetail deer and moose questionnaire were developed and distributed equally to hunters in both samples. The base scenario (zone A) was the same for all three versions and was intended to

Table 2. Results of binomial logit analyses used to estimate the probability of Saskatchewan big-game hunters accepting a higher cost to select a hunting site with different attributes.

Variable	Whitetail deer hunters				Moose hunters			
	Coefficient	SE	t-ratio	p	Coefficient	SE	t-ratio	p
CONSTANT	-0.580 23	0.2283	-2.542	0.011	-1.070 50	0.2395	-4.470	0.000
ACCESS	0.152 49	0.1266	1.205	0.228	0.281 65	0.1310	2.150	0.032
GAME	0.885 22	0.1344	6.585	0.000	1.317 20	0.1446	9.107	0.000
CONGEST	-0.735 63	0.1072	-6.864	0.000	-0.726 35	0.1119	-6.493	0.000
COST	-0.019 23	0.0038	-5.055	0.000	-0.008 76	0.0038	-2.307	0.021
EFFECTB	0.716 50	0.1121	6.390	0.000	0.704 26	0.1353	5.207	0.000
EFFECTK	0.511 70	0.1312	3.901	0.000	0.430 28	0.1322	3.255	0.001
FACTORH	-0.622 19	0.1078	-5.769	0.000	-0.207 70	0.1182	-2.207	0.027
Log likelihood test, χ^2		223.886				201.718		
McFadden's pseudo- R^2		0.099				0.098		

represent a mixed forest in northwestern Saskatchewan with no apparent forestry operations. The characteristics of this zone were (i) forest trails that are passable in dry weather with a two-wheel drive vehicle, but in foul or wet weather access is difficult even with a four-wheel drive vehicle (limited access); (ii) on a typical hunting day there will be evidence of 6 to 10 whitetail deer or 2 moose (low game populations); and (iii) a hunting party will not encounter another hunting party (low congestion). The alternative scenario (zone B) was intended to represent the same area following logging operations. The levels of access, game populations, and hunter congestion were varied to give a variety of post-harvesting circumstances. Six post-harvesting scenarios were examined in the study. Table 1 shows the variations of hunting site characteristics for the three versions of the surveys. The post-harvesting scenario always involved an increase in travel distance as a cost for the respondent. The cost terms in each survey were randomly generated using a uniform distribution, bounded by \$1.00 and \$50.00 as determined in pretests with actual hunters.

Contingent behaviour model development

Following Ben-Akiva and Lerman's (1985) notation, a new vector of attributes, x , is defined, which includes both z_{in} and S_n ; $x_{in} = h(z_{in}, S_n)$, and the indirect utility function, V_{in} , is now defined as $V_{in} = V(x_{in})$. This function is assumed linear in the parameters and with the β 's representing unknown parameters for K independent variables, the functions corresponding to zone A and zone B are

$$[13] \quad V_{0n} = \beta_1 x_{0n1} + \beta_2 x_{0n2} + \dots + \beta_K x_{0nK}$$

$$V_{1n} = \beta_1 x_{1n1} + \beta_2 x_{1n2} + \dots + \beta_K x_{1nK}$$

where 0 denotes zone A and 1 denotes zone B.

The final step in the specification of the binary choice model is the selection of the variables for inclusion in the indirect utility function. The initial variables chosen for the model were those that represented the differences in the attributes of the two zones: access, game populations, congestion, and travel cost. The data were set up to measure and record the differences in the attributes. These data were recorded as dummy variables taking on values of

0 for base or unimproved levels of quality and 1 for improved levels or quality (z_{in}).

In addition to the z_{in} , some individual specific characteristics of the hunters (S_n) were included in order to develop a model that accurately predicted the sites chosen. Many of these, such as income, hunting experience, and education, were not statistically significant. The final set of variables included in the analysis was as follows:

ACCESS	A dummy variable representing the quality of access to the hunting zone.
GAME	A dummy variable representing the expected game populations in the area.
CONGEST	A dummy variable representing the level of encountering other hunters in the area.
COST	The cost associated with the increased travel distance to zone B relative to zone A.
EFFECTB	A dummy variable representing how road access to new sites affects the respondent's hunting enjoyment. This variable was created from the 1-5 rating scale from the survey data. If road access to new sites decreases (or does not affect) hunting enjoyment (1, 2, 3), then EFFECTB = 0, and if it increases hunting enjoyment (4, 5), then EFFECTB = 1.
EFFECTK	A dummy variable representing how seeing a previously logged area replanted affects the respondent's hunting enjoyment. This variable was created from the 1-5 rating scale from the survey data. If seeing a previously logged area replanted decreases (or does not affect) hunting enjoyment (1, 2, 3) then EFFECTK = 0, and if it increases hunting enjoyment (4, 5), then EFFECTK became 1.
FACTORH	This variable is a dummy variable representing the effect of encountering other hunters to a respondent's hunting enjoyment. This variable was created from the 1-5 rating scale from the survey data. If these encounters decreased or did not affect the hunter's enjoyment (1, 2, 3), then FACTORH = 0, and if they increased enjoyment (4, 5), then FACTORH = 1.

Table 3. Estimated impacts (\$ per trip) of some forest environmental changes on the economic values of Saskatchewan big-game hunting trips.

	Scenario					
	1	2	3	4	5	6
Attributes						
Access	Limited	Improved	Improved	Improved	Limited	Improved
Game populations	Increased	Low	Low	Increased	Increased	Increased
Congestion	Increased	Increased	Low	Increased	Low	Low
Whitetail deer hunting trips						
Economic values						
Mean value ^a	31.96	17.79	32.03	35.75	53.01	58.22
Median value ^b	-8.53	-46.63	-8.38	-0.60	29.73	37.66
Standard deviation ^c	5.51	3.21	5.71	6.42	9.35	10.60
Moose hunting trips						
Economic values						
Mean value ^a	69.84	29.91	55.02	85.68	115.24	136.69
Median value ^b	-19.32	-137.47	-54.60	12.82	63.56	95.69
Standard deviation ^c	81.19	53.21	71.70	93.17	109.53	121.86

^aWhere $E(WTP)$ can only be greater than 0.^bWhere $E(WTP)$ can be less than or greater than 0.^cThese were derived from Monte Carlo simulations.

The binary logit model was estimated using maximum likelihood estimation techniques using LIMDEP (version 6.0) (Greene 1992). The number of individual respondents utilized in the estimation of the parameters were 741 deer hunters and 748 moose hunters.

Results

The results of the estimation procedures are shown in Table 2. The χ^2 statistics and associated significance levels show that the coefficients of both the deer and moose hunter models are significantly different from 0. The McFadden pseudo- R^2 value³ is 0.099 for the deer hunter model and 0.098 for the moose hunter model. Both models correctly predicted approximately 67% of the hunters' actual choices of site offered in the CB question.

In both models, the estimated coefficients of the parameters have the signs one would expect based on intuition and economic theory. For example, ACCESS and GAME are positive, indicating that an increase in access or game populations in zone B increases the probability that the hunter will choose to visit zone B. CONGEST and COST are negative, indicating that an increase in hunter congestion (decrease in privacy) in zone B or an increase in travel costs to get to zone B decreases the probability that the hunter will choose to visit zone B. EFFECTB is positive, indicating that if road access to new sites positively affects the respondent's enjoyment, then the probability that the hunter chooses to visit zone B will increase. EFFECTK is positive, indicating that if seeing previously logged areas

replanted increases hunting enjoyment, then the probability that the hunter will choose to visit zone B increases. This follows Wilman's finding that "vegetative characteristics that provide desirable habitat for game are likely to have some appeal for hunters" (Wilman 1984, p. 335). FACTORH is negative, indicating that if privacy is important in the selection of a hunting site, then the probability that the hunter chooses to visit zone B decreases.

In the deer hunter model all the parameters, except ACCESS, have t -statistics that indicate significance at the 99% level. Although ACCESS is insignificant, even at the 80% level, it remains in the models because it is one of the essential attributes being compared between zones A and B. In the moose hunter model all the parameters have t -statistics that indicate significance at the 95% level.

Welfare measures

The range of quality changes introduced in the CB questions give rise to six scenarios that form combinations of hunting site attributes. These are shown in the top portion of Table 3. To determine the impact of logging on the welfare of these Saskatchewan hunters, the scenario most closely resembling the expected post-harvesting situation must be determined. Stelfox's (1983, 1988) studies, combined with the habitat definitions given by Besko,⁴ suggest that it is likely that game populations increase in hunting zones where some clear-cut logging has taken place. Levels of the other quality variables can vary, however. If access to the area is also improved, hunters could be facing a hunting environment provided by scenarios 4 and 6, where

³ The calculation for McFadden's Pseudo- R^2 is as follows: $R^2 = 1 - (\log L \text{ of the unrestricted model} / \log L \text{ of the restricted model})$. For more information on McFadden's pseudo- R^2 see Maddala (1983).

⁴ Personal communication, M. Besko, ecologist, Terrestrial and Aquatic Environmental Management Ltd., northwestern Saskatchewan (April 1993).

improved access from forestry operations, increased game populations, and increased congestion (scenario 4) or low congestion (scenario 6) are presented. If access is not improved, hunters could face an environment provided by scenarios 1 and 5 where limited access, increased game populations, and increased congestion (scenario 1) or low congestion (scenario 5) are proposed.

Table 3 provides the mean (P^+) and median (P^-) WTP/trip, which were calculated using eqs. 11 and 12 for the six scenarios for deer and moose hunters. Values change substantially with each scenario. The mean WTP/trip associated with the environmental changes in scenario 1 (increased game populations, limited access, increased congestion) are \$31.96 for deer hunters and \$69.84 for moose hunters. The median WTP/trip (or $E(WTP)$ given that negative WTP values are admissible) is -\$8.53 for deer hunters and -\$19.32 for moose hunters. Scenario 2, in which access is improved, game populations are low, and congestion is increased, has lower mean WTP/trip at \$17.79 for deer hunters and \$29.91 for moose hunters. The median values for scenario 2 are -\$46.63 for deer hunters and -\$137.47 for moose hunters. The values associated with scenario 3, which has lower levels of congestion than scenario 2, suggest how decreasing congestion affects the welfare of the hunters: the mean WTP/trip for scenario 3 is \$32.03 for deer hunters and \$55.02 for moose hunters, while the medians are -\$8.38 and -\$54.60, respectively.

Scenario 4 has improved access, increased game populations, and increased congestion. The mean WTP/trip for this scenario is \$35.75 for deer hunters and \$85.68 for moose hunters. The median values are -\$0.60 for deer hunters and \$12.82 for moose hunters.

Scenario 5 is similar to scenario 1, except that congestion is decreased. A comparison of the welfare measures of scenario 5 and scenario 1 illustrate the effect of decreasing congestion when access is limited and game populations are increased. The mean WTP/trip for scenario 5 increases by \$21.05 to \$53.01 for deer hunters and by \$45.40 to \$115.24 for moose hunters; median values are \$29.73 for deer hunters and \$63.56 for moose hunters.

Scenario 6 has the most extensive environmental changes of the six different scenarios: improved access, increased game populations, and low congestion. Again the effects of decreasing congestion can be examined by comparing the welfare measures of scenario 6 and scenario 4. The mean WTP/trip for scenario 6 increases by \$22.47 to \$58.22 for deer hunters and by \$51.01 to \$136.69 for moose hunters. The median values are \$37.66 for deer hunters and \$95.69 for moose hunters.

Overall, the trip values are lower for deer hunters than for moose hunters. One reason for this may be that white-tail deer hunting in Saskatchewan is a local experience; hunters generally do not travel far to hunt whitetail deer. Therefore, the environmental changes offered in the various scenarios may not be significant enough to induce the deer hunters to travel extra distances to hunt.

The standard deviation of the WTP measures for the whitetail deer hunters is much smaller than the standard deviation of the mean WTP measures for the moose hunters. This suggests that there may be greater consensus among the whitetail deer hunters than the moose hunters regarding

the desirability of the environmental changes presented in the various scenarios.

The median WTP values are negative for many scenarios (Table 3). Negative medians imply that the changes outlined in those scenarios are not good enough to induce the majority of survey respondents to travel the hypothetical extra distance to zone B. If negative WTP values are admissible, the median WTP values suggest that deer hunters view scenarios 1-4 as reductions in welfare (relative to the base case), while scenarios 5 and 6 represent enhanced welfare situations. For moose hunters, scenarios 1-3 represent reductions in welfare, while scenarios 4-6 provide positive welfare effects.

The welfare measures described above reflect the value of changes in attributes as well as the fact that forestry operations bring about these changes. In each choice offered to the hunters, the "improved" site was characterized as having forest operations, while the "base" site had no forestry activity. It appears that forestry activity itself carries a negative value, as perceived by the hunters. This is evident from the negative intercept in the models (Table 2) and the fact that overall median WTP is still negative in cases where the "improved" site contains only changes in attributes with positive marginal utilities (e.g., scenario 3). This suggests that in some cases, the presence of forestry activity outweighs the positive changes in hunting attributes. In cases where both game populations and access are improved (scenarios 5 and 6), mean and median WTP are both positive and are more similar to each other than in the other scenarios. However, the median is still lower than the mean in these cases, indicating a skewed distribution of willingness to pay. In these cases, the attribute improvements outweigh the perceived negative effects from forestry activity.

The marginal values of each attribute per trip for deer hunters, holding all other factors constant, are \$7.93 for access, \$46.03 for game populations, and -\$38.25 for congestion. For moose hunters these values are \$32.15 for access, \$150.36 for game populations, and -\$82.92 for congestion. Positive marginal values for an attribute suggest that hunters become better off by positive increases in the levels of those attributes, while negative values suggest they become better off by decreases in the levels of those attributes. Clearly, moose hunters are more positively affected than deer hunters by improvements in game population and access. They are also significantly more negatively affected by congestion.

An application of the model: simulation of the impacts of logging

To illustrate an applied use of the model, the welfare measures were used to examine the values of changes in hunting environments in a particular area of Saskatchewan. Using information on logging-wildlife interactions (Stelfox 1983, 1988; Besko, see footnote 4), and discussing various landscape alterations with foresters, a simulation of the impact of post-harvesting conditions on the welfare of hunters was created for a given hunting management zone, zone 69, within the Norsask Forest Management License Area (FMLA) in the northwestern part of the province. The base-level attributes are assumed to correspond to

Table 4. Annual impacts of some forest environmental changes on the economic values of big-game hunting in zone 69, northwestern Saskatchewan.

	Scenario					
	1	2	3	4	5	6
Attribute						
Access	Limited	Improved	Improved	Improved	Limited	Improved
Game populations	Increased	Low	Low	Increased	Increased	Increased
Congestion	Increased	Increased	Low	Increased	Low	Low
Whitetail deer hunters						
Economic values						
Mean annual WTP ^a	\$10 419	\$5 800	\$10 442	\$11 655	\$17 281	\$18 980
Median annual WTP ^b	\$-2 781	\$-15 201	\$-2 732	\$-195	\$9 692	\$12 277
Moose hunters						
Economic values						
Mean annual WTP ^a	\$9 917	\$4 247	\$7 812	\$12 167	\$16 364	\$19 410
Median annual WTP ^b	\$-2 743	\$-19 521	\$-7 753	\$1 820	\$9 026	\$13 588

^aWhere $E(WTP)$ can only be greater than 0.^bWhere $E(WTP)$ can be less than or greater than 0.

current (pre-logging) conditions in northeastern Saskatchewan. Depending on the type of logging that takes place, and resulting changes in game populations, access, and congestion, the post-harvest condition may reflect one of the combinations of attributes described in scenarios 1–6. For example, it has been reported that 2- to 5-year-old cut blocks can provide excellent habitat for moose and deer. Thus these cut blocks represent an improvement in game populations. This characteristic, coupled with improved access due to logging roads, and no changes in congestion, would lead to scenario 6. The other scenarios may arise from other forest management strategies and (or) changes in administrative or social factors.

To apply the model and data to zone 69 a number of assumptions must be made. First, the number of trips by hunters to the zone is required. This information was collected from the hunters who completed the surveys. However, it was necessary to assume that these trips were taken for the purpose of hunting either deer or moose.⁵ This distribution of trips to various zones reported by the survey respondents was expanded to the entire provincial hunting population. Using this distribution, an estimated 326 trips were taken to zone 69 by deer hunters and 142 by moose hunters.

Second, post-timber harvesting effects on environmental quality were assumed to not influence the total number of trips to zone 69. This means that the changes due to forestry activity are not severe enough to influence hunters currently using the zone to hunt more, do not encourage individuals hunting in other areas of the province to switch to zone 69, and do not induce nonhunters to take up hunting. Without these assumptions, a more complex model

incorporating substitution of trips among zones would be needed to estimate changes in the trips to zone 69. Substitution effects that increase congestion faced by hunters currently active in zone 69, however, can be examined using the model.

Third, the quality of hunting in the other zones in the province is assumed to remain constant.

With these assumptions, welfare impacts were estimated by multiplying the number of trips to zone 69 by the per-trip welfare measure reflected by the various post-harvesting scenario. The resulting values, shown in Table 4, represent the aggregate annual value of the benefits hunters receive calculated using mean and median values. Scenario 6, with improved access, increased game populations, and no increase in congestion, and scenario 5, with limited access, increased game populations, and low congestion levels, provide the highest increase in welfare for both deer and moose hunters in the zone. Benefits decrease when congestion in the zone increases. These reduced benefit levels highlight the importance of privacy (low congestion) to these hunters. Not surprisingly, the benefits to hunters from limited levels of access are lower than the scenarios with improved access. Again, these benefits decrease when congestion in the region increases.

Assuming that these changes in values accrue to the hunters every year in perpetuity because of the environment provided by a regulated forest, and that there are no additional costs or benefits in future years, these annual values can be capitalized. Using a discount rate of 5%, which is consistent with current practice in the resource economics literature, the capitalized values for deer and moose hunters for increasing game populations and access levels while maintaining low levels of congestion are about \$380 000 evaluated at the mean and about \$270 000 at the median in zone 69. Note that these calculations, however, are based on assumptions of constant forest and socio-economic conditions. A more thorough analysis of this

⁵ While sample frame only included deer and moose hunters, these individuals reported hunting trips to zones that were not tied to specific species. In other words, some of their reported trips may have been taken for hunting other game for which they had purchased licences.

issue would involve simulation of changing forest and socioeconomic effects.

The capitalized values may be useful because they represent the benefits of maintaining the forest environments proposed in the six different scenarios. These benefits could be compared with the capitalized costs of changing harvesting practices to create or maintain such hunting environments. Thus, these values would be useful to the forest products company that holds the FMLA in zone 69 if it wished to compare various forest management plans which provided, for example, different qualities of game habitat or road access. The company could compare the costs of different management plans and associated timber values with the changes in hunting benefits and thus move closer to an IRM approach that considers impacts on a wider range of societal benefits. Alternative mechanisms that attempt to account for social benefits include regulation, tenure definitions, and the use of incentives. These issues are discussed elsewhere (e.g., Pearse 1993).

Summary and conclusions

This study develops an economic model that should be of considerable use in IRM. The model extends traditional CV analysis by evaluating multiple quality changes simultaneously, and by utilizing the CB framework. Most CV studies evaluated the values of environmental changes one at a time; evaluating multiple changes thus required multiple CV studies (notable exceptions include Hoehn and Loomis 1993). Such a method is expensive for forest companies and government agencies conducting environmental and social impact assessments or planning IRM strategies if they are unsure of the expected changes to the environment from harvesting operations. Therefore, the flexibility of the methodology used in this study is of great advantage.

The benefits summarized in Table 4 represent only a small portion of recreational activities that occur in one zone in the Saskatchewan FMLA area. In addition to white-tail deer and moose hunting, other use values include fishing, camping, hiking, and boating. If a benefit-cost analysis of forestry operations in the FMLA is to be performed, the impacts on other recreation in the area should be determined as well. It is unclear whether the welfare impacts of forest operations on these other recreational activities would be positive or negative. Furthermore, other values of the forest, such as those associated with old-growth stands or pristine wilderness areas, should also be considered.

In conclusion, the results from this study provide a way to estimate some of the social values for nontimber components of Canadian forests. The utility of this method is demonstrated by estimating the values associated with potential environmental changes following timber harvesting for two activities occurring in one portion of an FMLA in Saskatchewan. It is argued that this knowledge is essential for a successful integrated forest management plan. The study illustrates how hunters trade off environmental quality attributes and how they respond to the introduction of forestry operations in northwestern Saskatchewan. The results presented in this paper suggest that improvements in the CV method that incorporate behavioural changes and simultaneous environmental quality changes may facilitate

better involvement of the general public concerns and values into integrated resource decision-making processes.

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